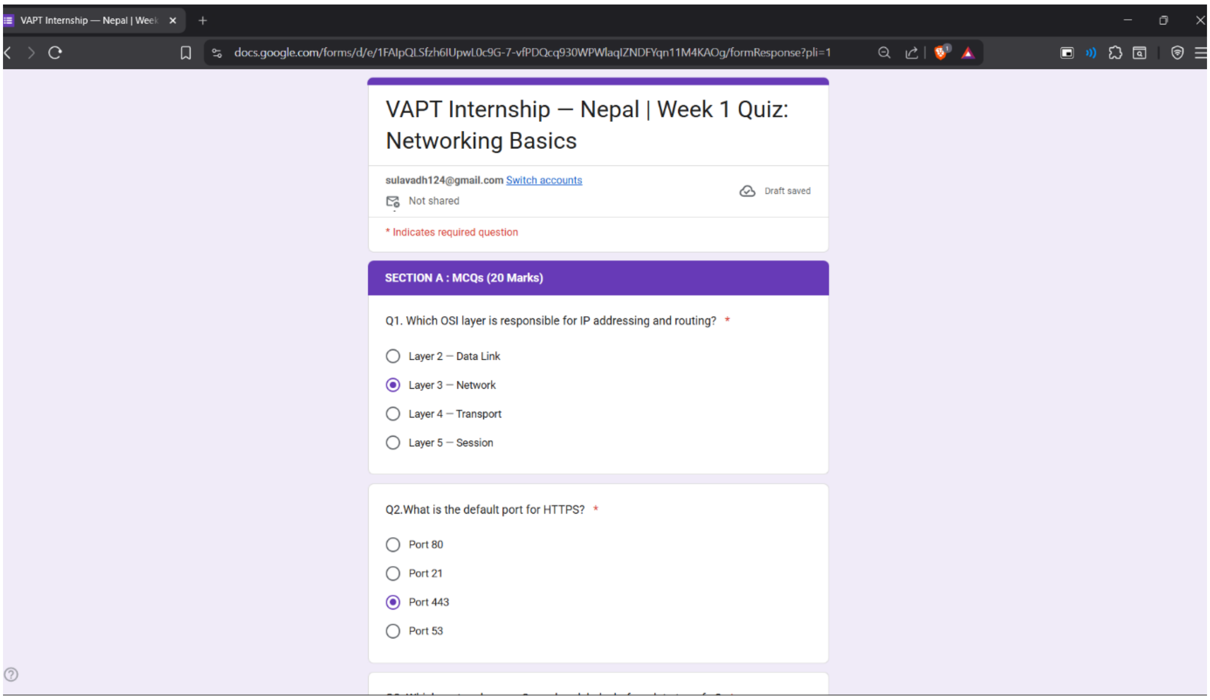
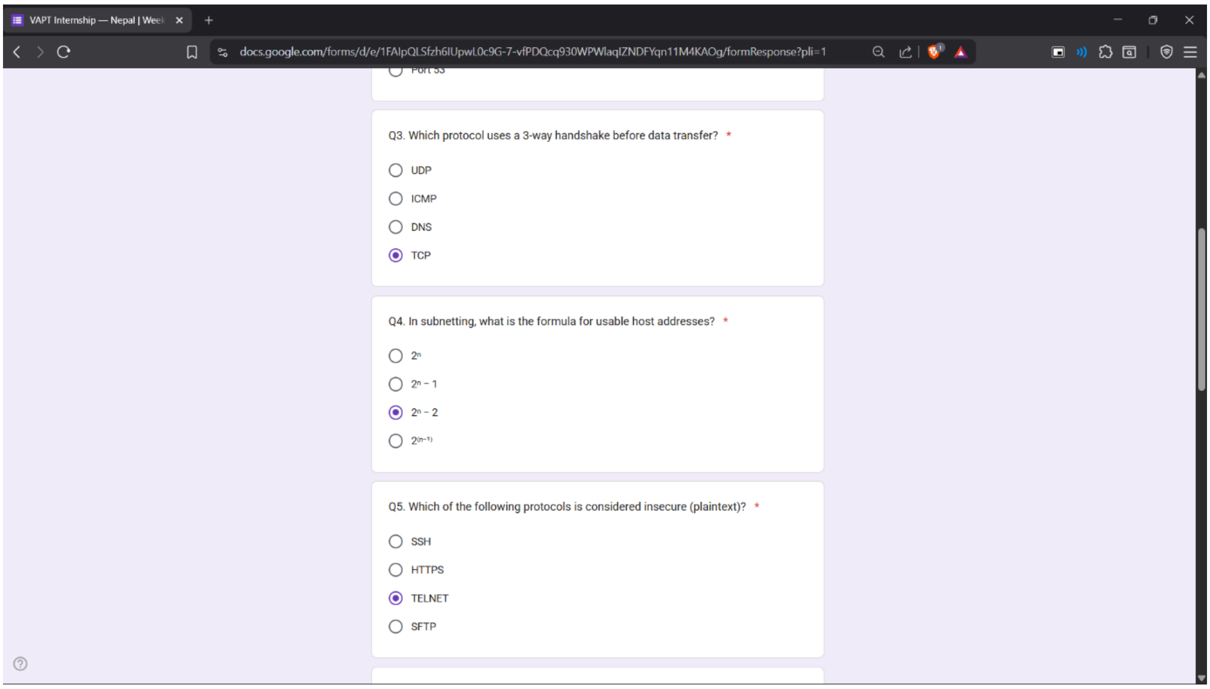
VPAT Internship

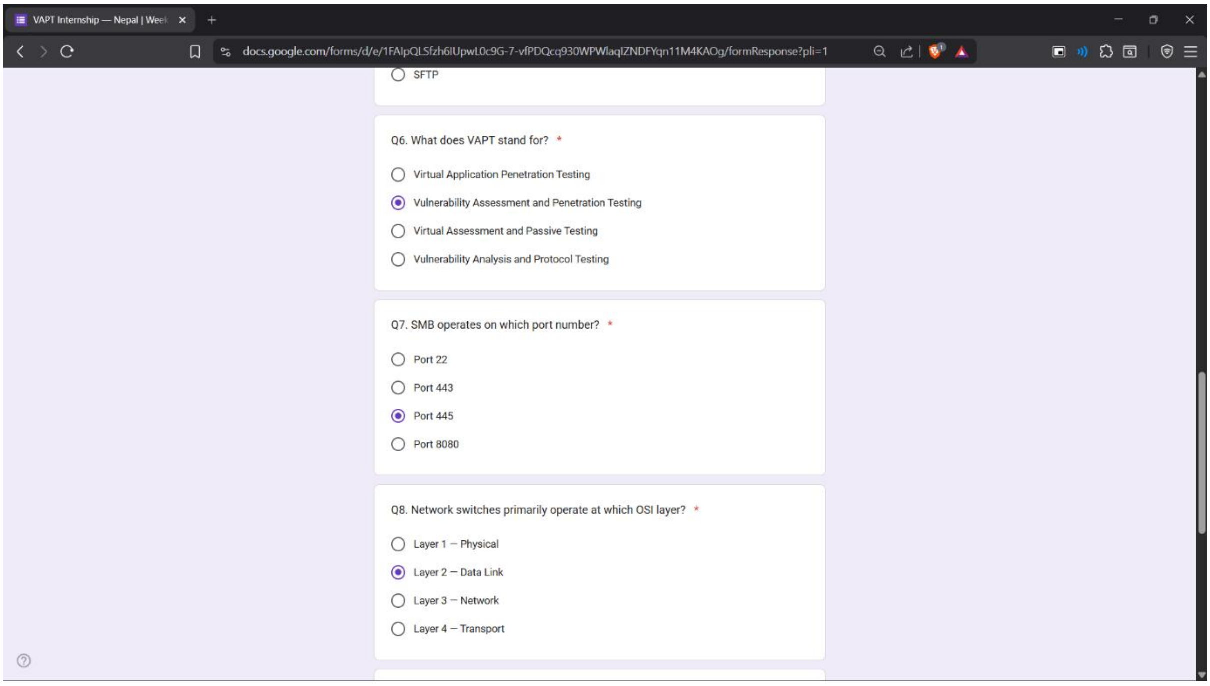
Sulav Adhikari

Week 1 Assignment

Section A:







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- ○ SFTP

Q6. What does VAPT stand for? *

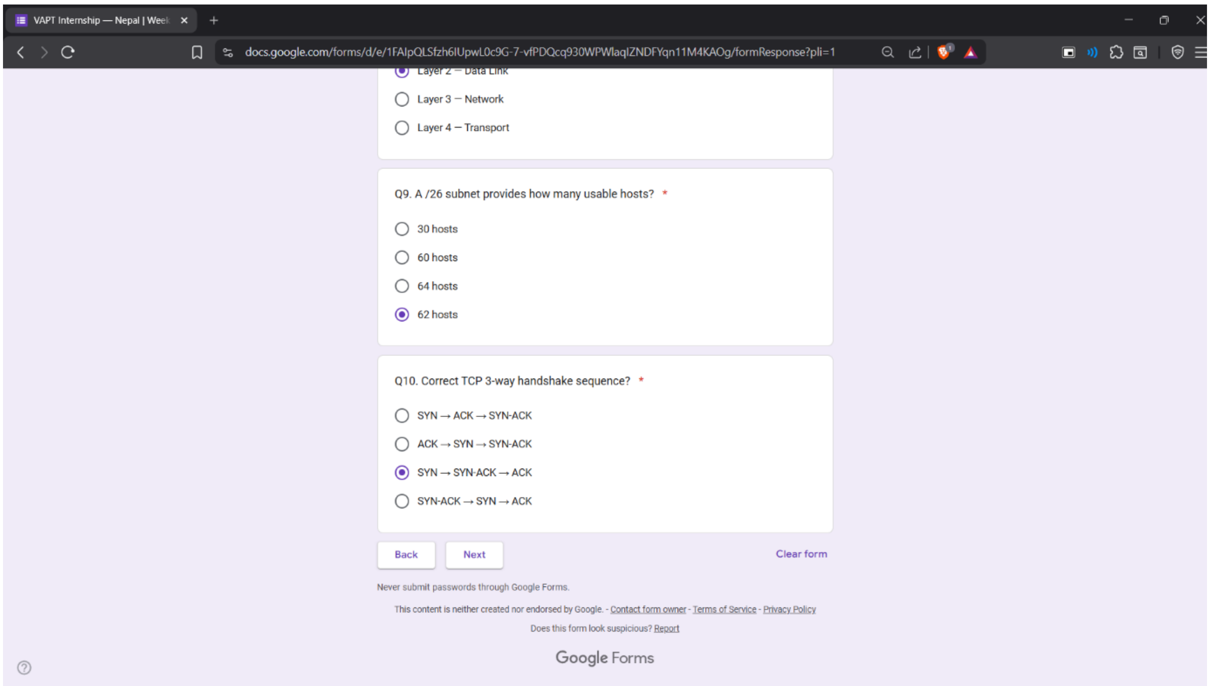
- ○ Virtual Application Penetration Testing
- ◉ Vulnerability Assessment and Penetration Testing
- ○ Virtual Assessment and Passive Testing
- ○ Vulnerability Analysis and Protocol Testing

Q7. SMB operates on which port number? *

- ○ Port 22
- ○ Port 443
- ◉ Port 445
- ○ Port 8080

Q8. Network switches primarily operate at which OSI layer? *

- ○ Layer 1 — Physical
- ◉ Layer 2 — Data Link
- ○ Layer 3 — Network
- ○ Layer 4 — Transport



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- ◉ Layer 2 — Data Link
- ○ Layer 3 — Network
- ○ Layer 4 — Transport

Q9. A /26 subnet provides how many usable hosts? *

- ○ 30 hosts
- ○ 60 hosts
- ○ 64 hosts
- ◉ 62 hosts

Q10. Correct TCP 3-way handshake sequence? *

- ○ SYN → ACK → SYN-ACK
- ○ ACK → SYN → SYN-ACK
- ◉ SYN → SYN-ACK → ACK
- ○ SYN-ACK → SYN → ACK

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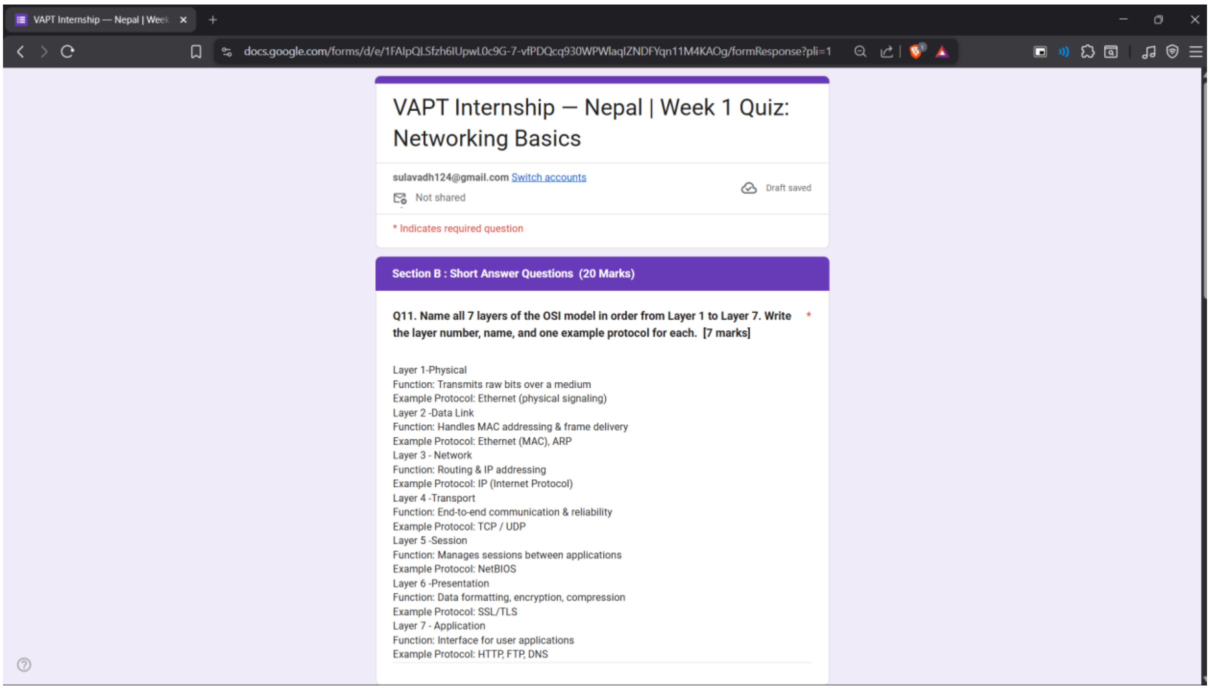
Section B:



VAPT Internship — Nepal | Week 1 Quiz: Networking Basics

sulavadh124@gmail.com Switch accounts

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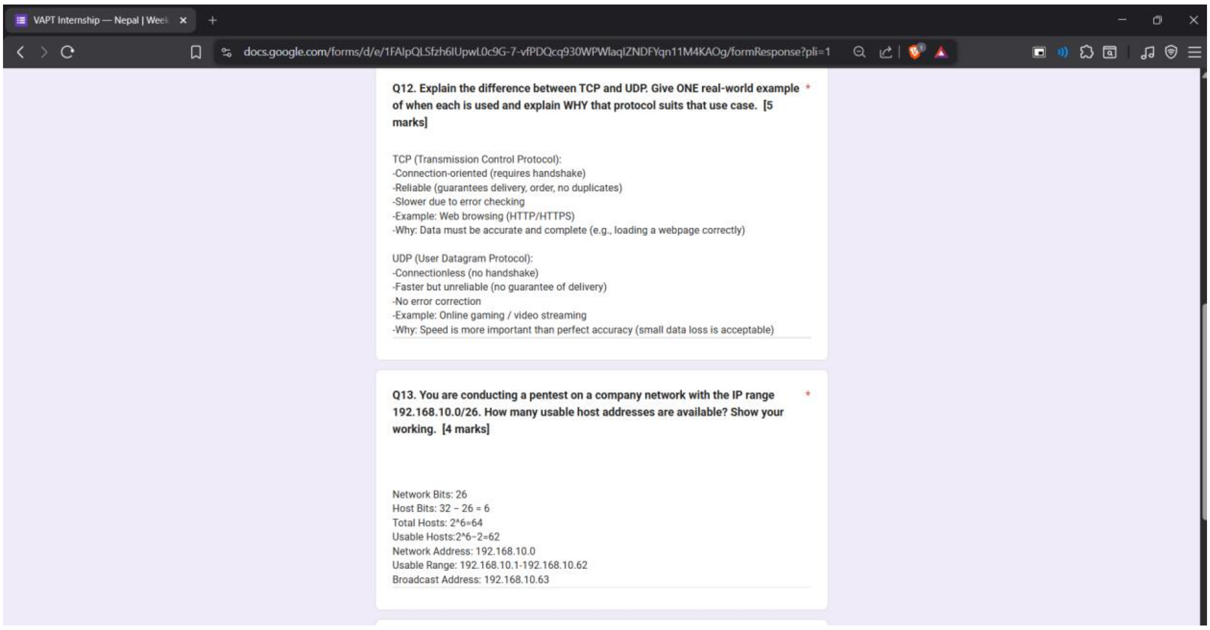
* Indicates required question

Section B : Short Answer Questions (20 Marks)

Q11. Name all 7 layers of the OSI model in order from Layer 1 to Layer 7. Write the layer number, name, and one example protocol for each. [7 marks] *

Layer 1-Physical
Function: Transmits raw bits over a medium
Example Protocol: Ethernet (physical signaling)
Layer 2 -Data Link
Function: Handles MAC addressing & frame delivery
Example Protocol: Ethernet (MAC), ARP
Layer 3 - Network
Function: Routing & IP addressing
Example Protocol: IP (Internet Protocol)
Layer 4 -Transport
Function: End-to-end communication & reliability
Example Protocol: TCP / UDP
Layer 5 -Session
Function: Manages sessions between applications
Example Protocol: NetBIOS
Layer 6 -Presentation
Function: Data formatting, encryption, compression
Example Protocol: SSL/TLS
Layer 7 - Application
Function: Interface for user applications
Example Protocol: HTTP, FTP, DNS



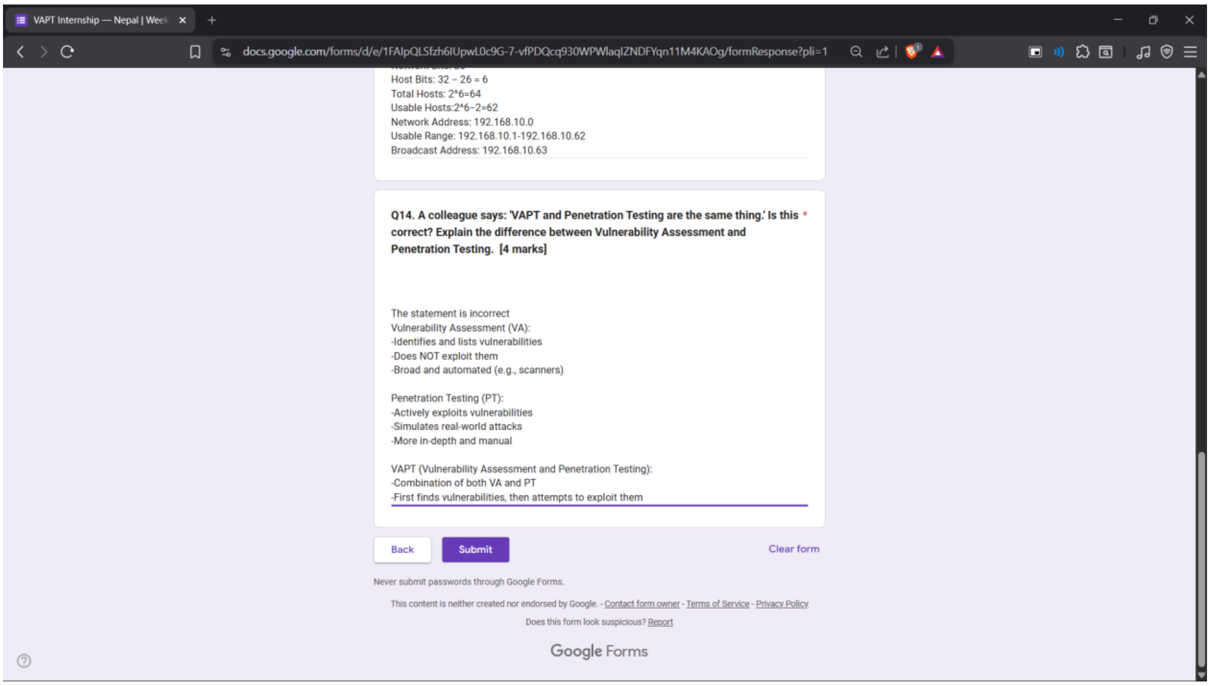
Q12. Explain the difference between TCP and UDP. Give ONE real-world example of when each is used and explain WHY that protocol suits that use case. [5 marks] *

TCP (Transmission Control Protocol):
-Connection-oriented (requires handshake)
-Reliable (guarantees delivery, order, no duplicates)
-Slower due to error checking
-Example: Web browsing (HTTP/HTTPS)
-Why: Data must be accurate and complete (e.g., loading a webpage correctly)

UDP (User Datagram Protocol):
-Connectionless (no handshake)
-Faster but unreliable (no guarantee of delivery)
-No error correction
-Example: Online gaming / video streaming
-Why: Speed is more important than perfect accuracy (small data loss is acceptable)

Q13. You are conducting a pentest on a company network with the IP range 192.168.10.0/26. How many usable host addresses are available? Show your working. [4 marks] *

Network Bits: 26
Host Bits: 32 - 26 = 6
Total Hosts: 2^6=64
Usable Hosts:2^6-2=62
Network Address: 192.168.10.0
Usable Range: 192.168.10.1-192.168.10.62
Broadcast Address: 192.168.10.63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Network Address: 192.168.10.0
Usable Range: 192.168.10.1-192.168.10.62
Broadcast Address: 192.168.10.63

Q14. A colleague says: 'VAPT and Penetration Testing are the same thing.' Is this correct? Explain the difference between Vulnerability Assessment and Penetration Testing. [4 marks] *

The statement is incorrect
Vulnerability Assessment (VA):
-Identifies and lists vulnerabilities
-Does NOT exploit them
-Broad and automated (e.g., scanners)

Penetration Testing (PT):
-Actively exploits vulnerabilities
-Simulates real-world attacks
-More in-depth and manual

VAPT (Vulnerability Assessment and Penetration Testing):
-Combination of both VA and PT
-First finds vulnerabilities, then attempts to exploit them

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Section C: Task C1: Protocol-Port Mapping Table

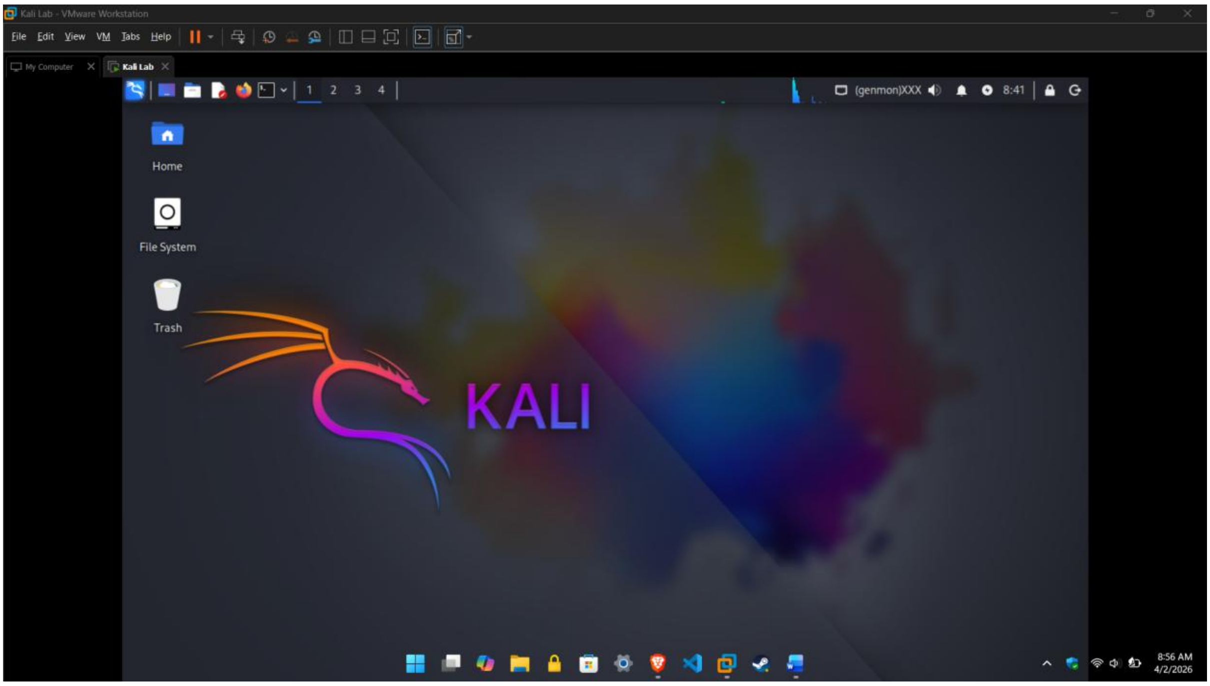
Protocol	Port	Transport (TCP/UDP)	Security Risk / Attack
HTTP	80	TCP	Data sent in plaintext, vulnerable to sniffing / MITM
HTTPS	443	TCP	SSL/TLS attacks (e.g., certificate spoofing)
FTP	21	TCP	Plaintext credentials, brute-force attacks
SSH	22	TCP	Brute-force login attempts
DNS	53	UDP/TCP	DNS spoofing / cache poisoning
RDP	3389	TCP	Brute-force attacks, unauthorized access
SMB	445	TCP	SMB exploits (e.g., ransomware like WannaCry)
Telnet	23	TCP	Plaintext communication, easy interception
SMTP	25	TCP	Email spoofing, spam attacks
DHCP	67/68	UDP	DHCP starvation / rogue DHCP server attacks

Task C2 - Subnet Calculation Worksheet

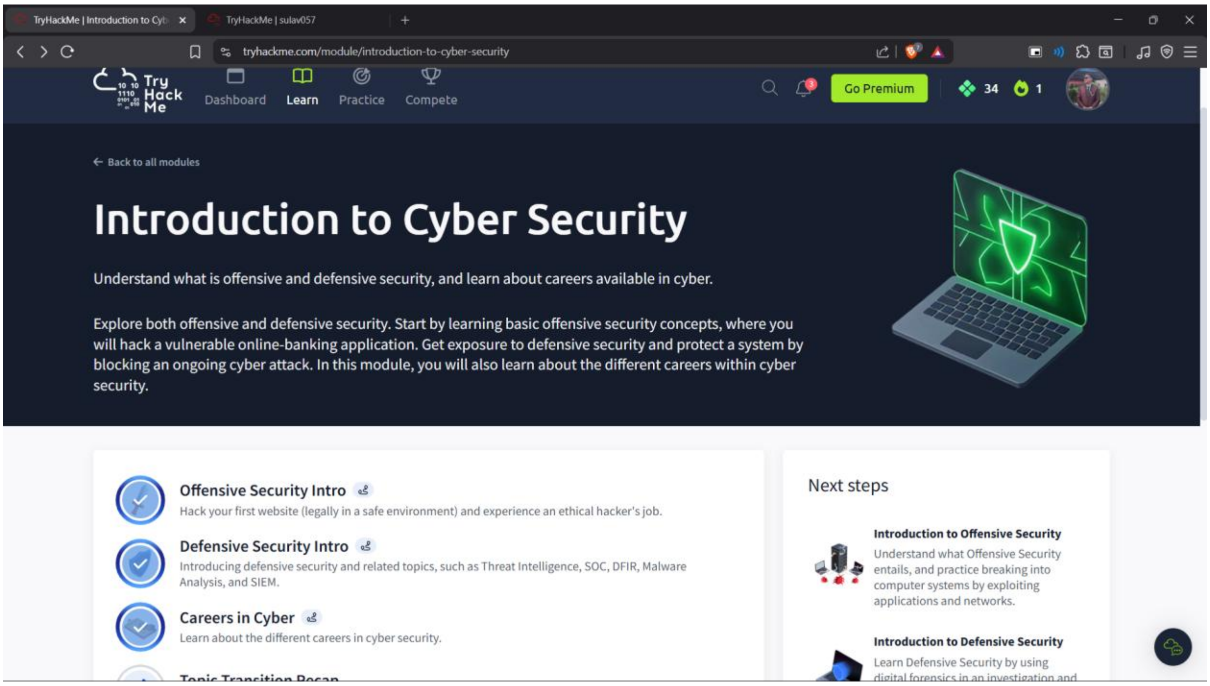
CIDR	Subnet Mask	Usable Hosts	Working
/24	255.255.255.0	254	$32-24=8 \rightarrow 2^8-2 = 256-2 = 254$
/25	255.255.255.128	126	$32-25=7 \rightarrow 2^7-2 = 128-2 = 126$
/26	255.255.255.192	62	$32-26=6 \rightarrow 2^6-2 = 64-2 = 62$
/27	255.255.255.224	30	$32-27=5 \rightarrow 2^5-2 = 32-2 = 30$
/28	255.255.255.240	14	$32-28=4 \rightarrow 2^4-2 = 16-2 = 14$
/30	255.255.255.252	2	$32-30=2 \rightarrow 2^2-2 = 4-2 = 2$

Task C3 - Screenshot Evidence

1. Kali Linux running on VirtualBox/VMware



2. TryHackMe: Pre-Security path



3. OSI Model